Project Proposal Guide (Weeks 7–8)

This worksheet guides the team in developing the Capstone/Software Engineering Project Proposal. It integrates the refined problem statement, findings from the Review of Related Literature (RRL), and the proposed solution concept.

Team Information

Project Title: **PRISM: Parenteral Return-demonstration Injection Skills Monitor**

Project Short Description: A mobile application that uses computer vision to detect insertion angle, aspiration technique, and withdrawal angle during nursing parenteral injection return demonstrations.

Team Code: 2526-sem2-it332-30

Members:

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PART 1: Introduction

Parenteral injection constitutes one of the most fundamental and safety-critical procedural competencies in nursing practice. Errors in needle insertion technique, particularly deviations in insertion angle and depth, are associated with adverse clinical outcomes including tissue trauma, medication infiltration, abscess formation, and peripheral nerve injury (Perry et al., 2020). In recognition of these risks, nursing education programs require students to demonstrate safe and accurate injection techniques prior to engagement in live clinical environments. At the Cebu Institute of Technology – University (CIT-U), procedural competence across the four major parenteral injection types, namely intramuscular, subcutaneous, intravenous, and intradermal, is formally assessed through return demonstrations wherein clinical instructors evaluate student performance against standardized rubric-based checklists.

However, rubric-based evaluation does not eliminate the influence of human judgment on assessment outcomes. Chen et al. (2023) established that clinical instructors frequently draw upon personal perceptions of student attitude and familiarity when scoring return demonstrations, resulting in divergent grading outcomes across evaluators. This is particularly evident in the assessment of injection execution steps, specifically needle insertion angle, aspiration technique, and needle withdrawal angle, which are objectively measurable yet consistently vulnerable to positional bias. Walsh et al. (2021) demonstrated that human observers may misjudge insertion angles by as much as 10 to 15 degrees due to parallax effects, while Tantacharoenrat and Precharattana (2024) identified angle determination as the foremost self-reported challenge in injection learning among nursing students and registered nurses. These evaluation limitations directly affect students' clinical self-efficacy and procedural confidence (George et al., 2020), and the demands of manually scoring technical steps across multiple students further compound instructor workload and elevate the risk of scoring error (Rodger & Juckes, 2021).

Although structured evaluation frameworks have been shown to improve inter-rater reliability in clinical competency assessment (Yu et al., 2021), current practices in nursing education continue to rely on unassisted manual observation with no mechanism for objective measurement of procedural execution. The absence of automated evaluation for insertion angle, aspiration, and withdrawal angle leaves both students and instructors without reliable performance data precisely where human judgment is most fallible. To address this problem, the proposed system is a mobile Android application that uses MediaPipe-based computer vision and the Llama 3.3 via Open Router API to detect and evaluate these three procedural components, providing partial objective insights that reduce instructor scoring burden and deliver immediate, data-driven feedback to nursing students at CIT-U.

PART 2: Objectives

2.1 General Objectives

This project aims to achieve the following general objectives within the second semester of Academic Year 2025–2026:

1. To design and develop PRISM (Parenteral Return-demonstration Injection Skills Monitor), a mobile Android application built using Flutter that utilizes the smartphone camera and MediaPipe-based computer vision to detect needle insertion angle, aspiration technique, and needle withdrawal angle across the four parenteral injection types, namely intramuscular (90°), subcutaneous (45°), intravenous (15°), and intradermal (10°), during nursing students' parenteral injection return demonstrations at CIT-U, targeting a measurement accuracy within three degrees of expert-annotated video baselines.
2. To integrate Llama 3.3 via the OpenRouter API within PRISM to process the compiled detection results of the three procedural components and generate structured, clinically worded performance feedback that is made available to users upon completion of each return demonstration session.
3. To design and implement a role-based module within PRISM that enables nursing students to access AI-generated feedback accompanied by targeted guidance for skill improvement, with all session records systematically stored in Firebase Firestore.

4. To develop a feedback release module within PRISM that enables clinical instructors to manage and distribute AI-generated performance feedback to nursing students through a Firebase Firestore-backed interface.

2.2 Specific Objectives

1.1 To develop a real-time camera module within the Flutter application that accesses the smartphone camera and integrates MediaPipe-based computer vision to detect and compute needle insertion angle across four parenteral injection types (intramuscular, subcutaneous, intravenous, and intradermal), targeting a measurement accuracy within three degrees of expert-annotated video baselines.

1.2 To implement a user input module within PRISM that allows users to select the type of parenteral injection (intramuscular, subcutaneous, intravenous, or intradermal) prior to demonstration, enabling appropriate configuration of system detection parameters based on the selected procedure.

1.3 To implement a detection module that processes smartphone camera input through MediaPipe to identify aspiration technique by recognizing plunger withdrawal motion and classifying whether it is correctly performed during the injection procedure, while measuring the accuracy of execution in terms of motion control and the duration of plunger withdrawal to assess procedural correctness.

1.4 To implement a detection module that measures the needle withdrawal angle and evaluates whether it corresponds to the insertion angle, in accordance with procedural standards defined in the rubric checklist.

2.1 To integrate Llama 3.3 via the OpenRouter API to process detection results from the evaluated procedural components and generate clinically worded feedback and performance scores that are displayed immediately after each return demonstration session.

3.1 To configure Firebase Authentication with role-based access control distinguishing nursing students and clinical instructors, ensuring appropriate access to session data within the application.

3.2 To implement a Firebase Firestore-backed data module that stores and displays AI-generated feedback and performance scores for students, while enabling instructors to monitor and review student performance records over time.

3.3 To conduct user evaluation testing with a minimum of five nursing students and two clinical instructors to assess the system's usability, functional performance, and effectiveness of AI-generated feedback interpretation, and to document user feedback and observed issues for system refinement and evaluation.

4.1 To implement a feedback review module that enables clinical instructors to review, refine, and release AI-generated feedback to nursing students through a Firebase Firestore-backed interface.

2.3 Research Questions

1. How accurately does PRISM measure needle insertion angle across the four parenteral injection types using the smartphone camera compared to an expert-annotated video baseline, and does it achieve a mean absolute error within three degrees?
2. To what extent does PRISM correctly detect aspiration technique and needle withdrawal angle during standardized return demonstrations, and how do these results compare with instructor evaluation?

PART 3: Methods

PRISM (Parenteral Return-demonstration Injection Skills Monitor) is an Android-based mobile application developed using the Flutter framework designed to mitigate human positional bias and cognitive load during nursing return demonstrations at the Cebu Institute of Technology – University (CIT-U). Addressing the unreliability of manual visual assessment, the system integrates MediaPipe-based computer vision and Llama 3.3 via OpenRouter API to provide objective measurement of three safety-critical components: needle insertion angle, aspiration technique, and needle withdrawal angle. The application caters to two primary stakeholders: nursing students, who select their injection type (IM, SubQ, IV, or ID) via a user-input module to receive immediate, clinically worded feedback, and clinical instructors, who utilize a role-based interface to monitor and review performance records. By utilizing vector-based geometric analysis—calculating the angle between the wrist (Landmark 0) and index fingertip (Landmark 8) relative to the forearm—PRISM quantifies performance against the CIT-U five-point rubric. Aspiration is detected through thumb-tip displacement tracking to confirm plunger withdrawal, while withdrawal angles are measured to ensure correspondence with the initial insertion. This approach digitizes the CIT-U procedural checklist into a data-driven tool, storing session records in Firebase Firestore without retaining sensitive video data, ensuring both pedagogical accuracy and data privacy.

The development of PRISM follows a Module-Based Development approach, where system components are engineered sequentially according to the functional dependencies outlined in the project objectives. This structured methodology ensures that foundational data-gathering modules are validated before the integration of higher-level analytical features. Development is divided into four primary stages:

1. **Core Detection Module:** Implementation of the real-time camera interface and MediaPipe integration to establish the geometric coordinate system for angle computation and motion tracking.

2. **User Logic & Authentication Module:** Development of the user-selection interface for injection types and the Firebase Authentication layer to manage role-based access for students and instructors.
3. **AI Integration & Feedback Module:** Configuration of the OpenRouter API to send detection parameters to the Llama 3.3 model for the generation of structured clinical feedback.
4. **Data Persistence Module:** Finalization of the Firebase Firestore schema to ensure all performance scores and AI-generated insights are systematically recorded and retrievable for longitudinal review.

The validation of PRISM focuses on technical precision and functional effectiveness, operating under the assumption of a 90% baseline for system usability. Technical accuracy is validated by measuring the system's detection outputs against a dataset of expert-annotated video baselines provided by CIT-U instructors. The primary metric is the Mean Absolute Error (MAE); the system is considered successful if it achieves an accuracy within three degrees for insertion and withdrawal angles. Aspiration detection is validated via binary classification accuracy (detected vs. not detected) compared to instructor ground truth. Following technical validation, functional performance is assessed through a pilot implementation in the CIT-U skills laboratory involving five nursing students and two clinical instructors. This phase measures the Agreement Rate between PRISM-generated scores and manual instructor evaluations, as well as the Reduction in Grading Time (seconds per student). By achieving these technical benchmarks, the system demonstrates its capacity to improve the consistency and efficiency of clinical competency assessments without the need for additional usability instrumentation.

PART 4: Expected System

The proposed system is an Android-based mobile application designed to provide an objective assessment and feedback platform for nursing students performing parenteral injection return demonstrations. The system aims to bridge the gap between traditional manual observation and expensive sensor-based equipment by utilizing widely available smartphone technology to capture and analyze procedural performance.

4.1 Key Features (Minimum Viable Product)

The Minimum Viable Product (MVP) of PRISM prioritizes the core technical functionalities designed to mitigate the documented inconsistencies inherent in human visual observation during clinical assessments.

1. **Configurable Injection Module:** A student-facing input module that enables the selection of four parenteral injection types—intramuscular (90°), subcutaneous (45°), intravenous (15°), and intradermal (10°)—which dynamically adjusts system detection parameters and success thresholds according to CIT-U standards.

2. **Real-time Computer Vision Detection:** A camera-based module utilizing the MediaPipe Hand Landmarker to compute needle insertion and withdrawal angles, targeting a mean absolute error within three degrees of expert-annotated video baselines.
3. **Advanced Aspiration Analysis:** A detection engine that recognizes thumb tip displacement to identify plunger withdrawal motion, evaluating the correctness of the technique in terms of motion control and duration to assess procedural proficiency.
4. **AI-Generated Clinical Feedback:** Integration with Llama 3.3 via the OpenRouter API to process compiled detection results and generate structured, clinically worded performance reports and improvement guidance within seconds of session completion.
5. **Instructor Decision-Support Dashboard:** A role-based interface designed for clinical instructors to monitor student progress, review AI-generated scoring metrics, and refine feedback based on professional judgment prior to its final release.
6. **Cloud-Based Record Management:** A Firebase Firestore-backed data module providing secure storage for session records and performance scores, governed by role-based authentication to ensure data privacy and integrity.

4.2 High-Level Workflow

The conceptual workflow of PRISM is structured across three sequential phases: Input/Configuration, Real-time Processing, and Validated Output.

1. **Authentication and Configuration:** Users authenticate via Firebase, which directs them to role-specific interfaces for students or instructors. Students initiate sessions by selecting a parenteral injection type, establishing the target geometric parameters and rubric-based success criteria.
2. **Real-time Capture and Analysis:** The student positions the device camera to record the return demonstration. The system processes the live feed using MediaPipe to compute hand landmarks, measuring insertion angle, monitoring aspiration motion control, and verifying withdrawal angle correspondence.
3. **AI Feedback Generation:** Following procedure completion, the system transmits detection data—including angular deviations and aspiration results—to Llama 3.3. The AI interprets these metrics against institutional standards to generate a structured, clinically worded performance summary.
4. **Review and Validation:** Session records are persisted in Firestore. Clinical instructors access the dashboard to review AI-suggested results, serving as the final authority to refine or validate the feedback for clinical accuracy before releasing the report for student review.

PART 5: Discussion

The scope of the proposed system is limited to the development of an Android-based mobile application that supports the objective assessment of selected measurable components of parenteral injection return demonstrations within a controlled educational environment.

Specifically, the system addresses three procedural variables that can be reliably detected and quantified through computer vision: needle insertion angle across four injection types (subcutaneous at 45°, intramuscular at 90°, intravenous at 15°, and intradermal at 10°), aspiration motion through dominant thumb tip landmark displacement following confirmed needle insertion, and withdrawal angle using the same vector computation applied at the moment of needle removal. Scoring thresholds are configurable and mapped directly to CIT-U's five-point performance scale per injection type, aligning system outputs with institutional rubric standards rather than generic benchmarks. The application is designed for use during skills laboratory sessions at CIT-U and is intended to supplement, rather than replace, instructor judgment by providing objective reference measurements for the three procedural steps where human observation is most susceptible to parallax error and evaluator subjectivity.

The system supports two primary user groups with distinct functional interfaces. Nursing students access a results view displaying insertion angle score, aspiration detection result, withdrawal angle score, and AI-generated clinical feedback immediately after each session. Clinical instructors access a role-based records view through Firebase Firestore displaying session scores and AI feedback per student, enabling asynchronous monitoring and review. Instructors retain final authority to refine or validate AI-generated feedback based on professional judgment prior to its release to students. AI feedback is generated by Llama 3.3 70B Instruct via the OpenRouter API, producing structured, clinically worded performance commentary from detection results within approximately two seconds of session completion.

Several limitations are anticipated. First, the system evaluates only externally observable movements captured by the device camera and cannot assess tactile components such as force application, needle stability within tissue, patient discomfort, or aseptic technique. Second, measurement accuracy may be influenced by environmental conditions including lighting quality, camera placement, background clutter, occlusion of the syringe or hand, and variations in smartphone hardware. Third, aspiration motion classification depends on the visibility and displacement of the dominant thumb tip landmark, which may introduce detection inconsistencies when hand positioning partially obstructs the relevant landmark. Reliable operation may also depend on stable internet connectivity for Firebase Authentication, Firestore data storage, and OpenRouter API calls for AI feedback generation. Additionally, because evaluation is confined to the three selected measurable variables, the system does not constitute a complete assessment of the entire injection procedure and must be used in conjunction with instructor judgment.

Despite these constraints, the expected contributions of the project are assessed through three research questions. RQ1 examines whether PRISM achieves a mean absolute error within three degrees for needle insertion angle measurement across all four injection types, validated against a dual-source expert-annotated video dataset combining internet instructional footage and CIT-U return demonstration recordings. RQ2 evaluates the accuracy of aspiration technique detection and withdrawal angle correspondence measurement against instructor-assigned ground truth. RQ3 assesses system usability through the System Usability Scale (SUS), targeting a score of 80 or higher, supplemented by qualitative feedback from a minimum of five nursing students and two clinical instructors during User Acceptance Testing at CIT-U.

For clinical instructors, the system functions as a decision-support tool that reduces scoring variability by supplementing subjective observation with objective angle and motion measurements, and reduces evaluation workload by centralizing student performance records accessible through a role-based Firebase Firestore interface. For nursing students, the availability of immediate, structured AI feedback is expected to improve awareness of procedural accuracy and support targeted skill refinement during practice sessions.

Overall, the project contributes to nursing education by demonstrating how mobile computer vision and AI-generated feedback can augment traditional skills assessment through measurable improvements in evaluation efficiency, scoring consistency, and feedback immediacy. By grounding its contribution in both technical performance validated across three research questions and quantifiable user impact assessed through standardized usability measures, the study provides a defensible and comprehensive basis for evaluating the role of emerging technologies in clinical competency education while preserving the essential role of instructors in holistic assessment.

PART 6: Traceability Matrix

RRL Finding / Theme	**Identified Gap**	**Research Question**	**Proposed Function**
Human observers misjudge injection angles by 10–15° due to parallax effects (Walsh et al., 2021)	No objective, camera-based angle measurement tool exists for nursing return demonstrations using consumer smartphone hardware	RQ1 — How accurately does PRISM measure needle insertion angle across four injection types compared to expert-annotated video ground truth?	Real-time insertion angle detection module using MediaPipe vector-based landmark computation for SubQ (45°), IM (90°), IV (15°), and ID (10°)

Smartphone-based angle guidance reduces median angular error from 9.0° to 2.7° compared to freehand estimation (Saccenti et al., 2025)	No validated marker-free angle assessment system for nursing procedures exists using only consumer smartphone hardware (Gap 2 — RRL worksheet)	RQ1 — angle accuracy within specified threshold	MediaPipe Hand Landmarker on-device detection — no external hardware or markers required
Motion trajectories from hand landmarks classify procedural steps with ~83% accuracy (Lee et al., 2020; Gautier et al., 2021)	No existing system detects aspiration motion during nursing injection return demonstrations using smartphone camera	RQ2 — To what extent does PRISM correctly detect aspiration technique and withdrawal angle?	Aspiration motion detection module based on dominant thumb tip landmark displacement following confirmed needle insertion
Withdrawal angle must match insertion angle per CIT-U rubric Step 22 — no automated system verifies this	No tool measures withdrawal angle correspondence during nursing injection return demonstrations	RQ2 — rubric step detection accuracy	Withdrawal angle measurement module applying same vector computation as insertion angle at moment of needle removal
Most AI assessment systems validated on lab-controlled footage — not real clinical training environments (Choi et al., 2026; Chang et al., 2025)	No AI assessment system validated against authentic institution-provided clinical footage in Southeast Asian undergraduate nursing context (Gap 1 — RRL worksheet)	RQ1 and RQ2 — accuracy validation methodology	Dual-source video validation dataset combining internet instructional footage and CI-provided return demonstration recordings from CIT-U

Simulation-based assessment only valid when rigorously standardized (Hayden et al., 2014)	No standardized technology-based assessment instrument aligned to CIT-U nursing competency standards currently exists	RQ3 — How do users rate PRISM usability?	Configurable scoring thresholds mapped directly to CIT-U five-point performance scale per injection type
AI-assisted evaluation improved objectivity and reduced instructor workload (Wang et al., 2025; Chang et al., 2025)	No integrated system combines smartphone angle detection with AI-generated clinical feedback for nursing education (Gap 4 — RRL worksheet)	RQ1 and RQ2 — System accuracy and feedback quality	Llama 3.3 70B Instruct via OpenRouter API generating structured clinical feedback from detection results within two seconds
Instructor subjectivity and rater variability documented across clinical evaluator studies (Faherty et al., 2020; Gormley et al., 2012)	No objective measurement data provided to instructors during or after nursing return demonstrations	RQ3 — User-rated usability and perceived utility of instructor dashboard	Role-based instructor records view displaying session scores and AI feedback per student via Firebase Firestore
Mobile video feedback disproportionately benefits low-performing students (Miao et al., 2025; Kaya & Durgun, 2025)	No immediate automated feedback available to nursing students after return demonstrations at CIT-U	RQ4 — feedback availability improvement	Student results view displaying insertion angle score, aspiration detection result, withdrawal angle score, and AI feedback immediately after session

PART 7: References

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